Deep Spectral Sensing and Reconstruction for High-Resolution Imaging of MWP-SAR in Complex Electromagnetic Environments

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***Abstract*— High-resolution synthetic aperture radar (HR-SAR) is extensively used in ground remote sensing applications, including disaster monitoring and resource crop assessment. This is attributed to its exceptional high-resolution imaging capabilities. Microwave photonics technology plays a crucial role in enhancing the performance of SAR systems. It enables the direct emission of ultra-wideband signals across multiple frequency bands. However, this advancement also makes microwave photonic synthetic aperture radar (MWP-SAR) susceptible to complex and diverse electromagnetic interference. Particularly, it is vulnerable to radio frequency interference (RFI), and this interference seriously affects the high-resolution imaging results. In order to solve these problems, an MWP-SAR imaging algorithm based on depth spectral sensing and spectral reconstruction that is suitable for complex electromagnetic environments is proposed. Simultaneously, a sensing-removing-recovering (SRR) anti-interference imaging framework is established. To address the high-precision detection of various unknown interferences in MWP-SAR echoes, a deep learning network is constructed. This network is based on spectrum sensing theory and achieves an interference detection probability greater than 98% in various interference scenarios. Subsequently, targeting the continuous spectrum loss in the MWP-SAR signal after interference removal, a signal reconstruction algorithm is proposed. This algorithm employs Toeplitz transformation for signal loss scenarios, facilitating the recovery of signals with extensive spectrum loss post-interference removal. The factor group sparse regularization (FGSR) algorithm is used to quickly solve the signal recovery problem. Through simulation and measured data processing, the superiority of the proposed algorithm over existing interference suppression imaging algorithms is demonstrated.**

***Index Terms*— Anti-interference, high-resolution imaging, matrix completion, microwave photonic synthetic aperture radar (MWP-SAR), spectrum sensing.**

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I. Introduction

HIGH-RESOLUTION synthetic aperture radar (HR-SAR) has been widely applied in disaster monitoring, marine protection, resource exploration, and topographic mapping due to its advantages of all-weather, all-day, and high resolution [1], [2]. With the advancement of microwave photonics technology, the transmission bandwidth of SAR has been significantly enhanced, enabling the emission of ultra-wideband signals across multiple bands, with an instantaneous working bandwidth exceeding 10 GHz [3], [4]. This greatly enhances the detection capability of SAR, making microwave photonic synthetic aperture radar (MWP-SAR) to have better performance and application in target detection, target identification, and port traffic monitoring than traditional SAR [5], [6], [7].

Ultra-wideband signals are prone to interference within the intricate spectral space of the operational frequency band [8]. Forms of interference include a variety of radio frequency interference (RFI), such as from wireless communications, television broadcasts, and other radar transmissions present in the electromagnetic spectrum [9], [10], [11]. There is also repeater jamming caused by dedicated jammers [12]. Consequently, despite the capability of SAR to achieve significant energy accumulation via 2-D matched filtering, the presence of robust interference sources significantly compromises the quality of imaging results. This interference profoundly impacts the detection efficacy of MWP-SAR, elevating the complexity of subsequent image interpretation.

This study directs its attention toward the extensive application domain of MWP-SAR within civilian contexts, specifically addressing high-resolution imaging challenges in complex electromagnetic environments. The principal impediment to achieving high-resolution imaging predominantly arises from formidable power suppression interference across temporal, spectral, and spatial domains [13], [14]. Consequently, this research primarily investigates the intricacies of high-resolution imaging in MWP-SAR amid frequency

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suppression interference. It turns out that it is particularly challenging to propose efficient anti-interference algorithms in complex electromagnetic environments with broadband RFI. Existing anti-interference algorithms can broadly be classified into three categories: non-parametric, parametric, and semi-parametric.

Parametric interference suppression algorithms necessitate the formulation of a signal model for the interference signal and subsequent estimation of its parameters. This involves the assessment of the frequency of isolated broadband interference using iterative adaptive approach (IAA) and short-time Fourier transform [15], [16], followed by interference suppression achieved through projection onto orthogonal subspaces. Djukanovic and Popovic [17] proposed an interference estimation method based on high-order fuzzy functions, using polynomials to model interference phase parameters. Nevertheless, these algorithms face two critical limitations. First, in the absence of prior knowledge, they struggle to accurately reconstruct the interference signal, resulting in diminished interference suppression performance. Second, due to the necessity of exploring high-dimensional parameters, there exists significant room for improvement in terms of computational efficiency. Additionally, parametric algorithms are generally designed for handling isolated interference in the echo and are ill-suited for scenarios involving multiple interferences that may be present in the echo of MWP-SAR.

Non-parametric interference suppression algorithms leverage the intensity characteristics of both interference and real signals to formulate adaptive filters for interference elimination [18]. These encompass methods such as narrowband notch filters, broadband notch filters, bidirectional notch filters, and image domain notch filters. Nevertheless, exceeding a null width of 2% in the notch filter compromises the spatial resolution of SAR [19], concurrently elevating waveform sidelobes. To address this challenge, a subspace projection method based on eigenvalue decomposition has been proposed, effectively segregating the signal space from the interference subspace [20]. Despite its efficacy, this method encounters limitations in handling low signal-to-noise ratio (SNR) echo signals and struggles with scenarios involving multiple interferences within the echo simultaneously. At the same time, Wu's team proposed a 2-D space-time filter method that can accurately remove interference [21]. However, this algorithm relies on the performance of orthogonal waveforms and has high system complexity.

Semi-parametric interference suppression algorithms leverage characteristics such as sparsity and low-rank properties within signals [22], [23]. They transform the signal separation challenge into an optimization problem with hyperparameters for resolution. In comparisons to parametric methods, these approaches mitigate the intricacies and complexities associated with multiparameter estimation. The application of tensor technology for interference suppression has demonstrated commendable efficacy against complex interference scenarios [24]. Nevertheless, the efficacy of this method hinges on the values of hyperparameters and the choice of optimization models. Despite these advancements, the MWP-SAR context introduces various interference types that are both diverse and time-varying. Consequently, achieving robust anti-interference performance in this dynamic environment remains a formidable challenge.

To address the aforementioned challenges, the anti-interference algorithm architecture illustrated in Fig. 1 is presented in this article. This anti-interference imaging process can be summarized as sensing-removing-recovering (SRR). Initially, the theory of spectrum sensing [25], [26] is introduced, and an interference depth detection network is established. This network learns the frequency-domain characteristics of MWP-SAR echoes, yielding optimal parameters for interference detection. Subsequently, diverse interference types within the echoes are promptly identified and eliminated through the online detection network. Simultaneously, to counteract the impact of removed signal spectra on imaging resolution, this study employs the factor group-sparse regularization (FGSR) algorithm to restore the missing signal spectrum. The proposed network autonomously detects and eliminates various interferences in MWP-SAR echo signals, achieving a remarkable 98% interference detection accuracy in the test set. Furthermore, the subsequent broadband spectrum recovery algorithm effectively restores high-quality MWP-SAR ultra-wideband frequency signals, even when the interference area reaches 60%. Finally, this article verifies the effectiveness of the algorithm by processing the measured MWP-SAR data.

The algorithm presented in this article exhibits two significant advantages in comparisons to traditional anti-interference algorithms.

1) The interference sensing network formulated in this study demonstrates exceptional accuracy in detecting various types of suppression interference, achieving a perception accuracy exceeding 98%.
2) Even in scenarios where a substantial portion of the MWP-SAR spectrum is subject to interference, this algorithm effectively preserves imaging resolution.

In summary, the article is organized as follows. Section II initiates the establishment of the signal model for MWP-SAR radar under interference. Section III details the designed SRR imaging architecture, which includes a dedicated spectrum sensing network and signal recovery algorithm. Section IV analyzes the learning performance of the established interference detection network in detail. Then, Section V gives the simulation and measured data processing results of the algorithm. Finally, conclusions are given in Section VI.

II. MWP-SAR Interference Echo Modeling

This section focuses on establishing the MWP-SAR echo model and analyzing diverse interferences within MWP-SAR echoes. By establishing the echo time-domain model in a complex electromagnetic environment, it provides a foundation for the establishment of subsequent SRR anti-interference imaging architecture.

A. MWP-SAR Echo Model

MWP-SAR exhibits a larger emission bandwidth and synthetic aperture compared to traditional SAR systems. The

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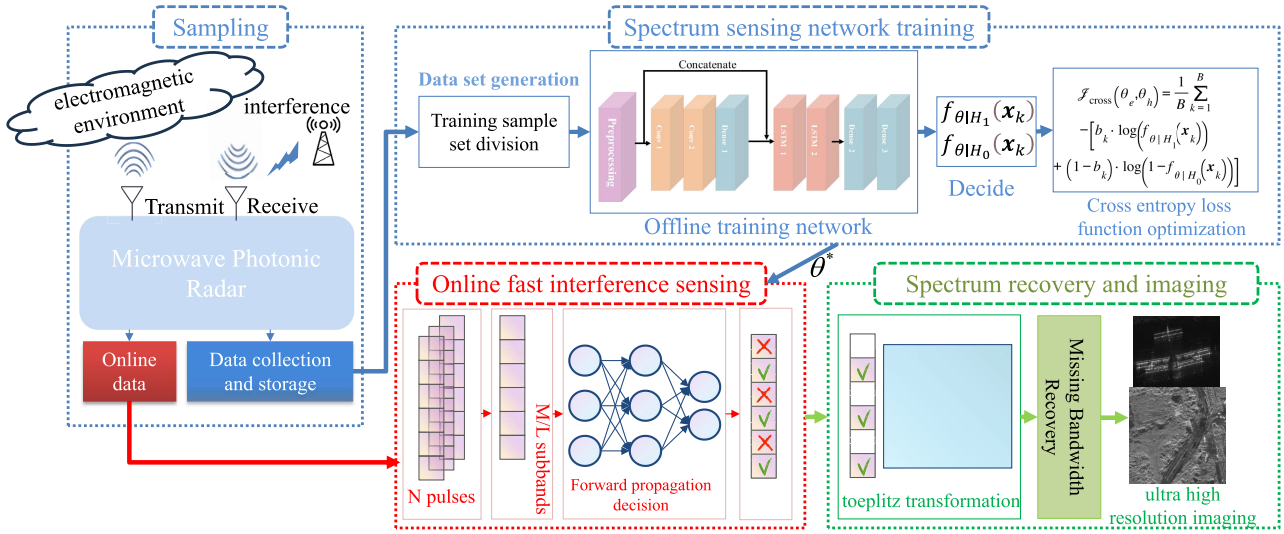


Fig. 1. MWP-SAR complex electromagnetic environment imaging architecture.

MWP-SAR echo has obvious reflection differences at different incident angles and frequencies [27], [28], which makes a huge difference between the MWP radar echo model and the ideal point target model. This article describes the reflection characteristics of microwave photonic radar echoes by introducing the widely accepted ASC model [29]. In contrast to the conventional echo model, the ASC echo model incorporates a frequency-dependent phase term and a reflection azimuth dependence term. Specifically, when the transmitted signal is a linear frequency modulation signal, the MWP-SAR echo modeled by the ASC model can be expressed as follows:

$$S_{\mathrm{ASC_}f}(\eta, f) = \sum_{i=1}^{I} A_i \cdot \left(j\frac{f}{f_c}\right)^{\alpha_i} \cdot \omega_r\left(\frac{f - f_c}{k_r}\right) \cdot \omega_a(\eta - \eta_{ci}) \cdot \exp\left[-j\pi\frac{(f - f_c)^2}{k_r} - j4\pi f\frac{R_i(\eta)}{c}\right]. \quad (1)$$

In this context, I signifies that the echo comprises I scattering points. α_i denotes the reflection coefficient of the ith reflection point, where $(j(f/f_c))^{\alpha_i}$ signifies the additional scattering phase of the ith scattering point. Here, η and f denote the azimuth time and range frequency, respectively. η_{ci} represents the azimuth imaging center, and $R_i(\eta)$ denotes the distance history of the ith scattering center. The modulation frequency of the transmitted signal is denoted by k_r, while c and f_c represent the speed of light and the center frequency of the emitted signal, respectively. $\omega_r(\cdot)$ and $\omega_a(\cdot)$ represent the rectangular windows of the echo signal in the range and azimuth directions, respectively.

Due to interference from other sources in a complex electromagnetic environment, radar signals are superimposed in the time domain. In order to measure the echo characteristics of the echo signal in the range time domain, this article gives the time-domain expression of the signal. According to the principle of stationary phase (POSP), the echo in the 2-D time domain can be expressed as

$$S_{\mathrm{ASC_}t}(\eta, t) = \sum_{i=1}^{I} A_i \left[1 + \frac{k_r}{f_c}\left(t - \frac{2R_i(\eta)}{c}\right)\right]^{a_i} \cdot \omega_r\left(t - \frac{2R_i(\eta)}{c}\right) \cdot \omega_a(\eta - \eta_{ci}) \cdot \exp\left\{j\pi k_r\left[t - \frac{2R_i(\eta)}{c}\right]^2 - j4\pi\frac{R_i(\eta)}{\lambda}\right\} \quad (2)$$

whereas t signifies the range sampling time in this context, the literature [5] has demonstrated the minimal impact of the scattering phase related to range time on the echo, as expressed in the formula. Even if it affects imaging to some extent, this aspect can be effectively compensated and corrected through the algorithm outlined in [6]. Therefore, this article excludes this component in subsequent derivations and analyses. At the same time, since the interference detection for each pulse signal is an independent processing process. Therefore, the scattering phase in the MWP-SAR echo along the azimuth direction can also be ignored. Considering only one pulse signal, the echo signal sampled in this direction can be expressed as

$$S_t(\eta, t) = \sum_{i=1}^{I} A_i \cdot \exp\left\{j\pi k_r\left[t - \frac{2R_i(\eta)}{c}\right]^2 - j4\pi\frac{R_i(\eta)}{\lambda}\right\}. \quad (3)$$

B. MWP-SAR Interference Model

Since the MWP-SAR's transmitted signal has a very wide signal frequency band, its receiving frequency band may contain a variety of interference signals, which greatly affect the ultrahigh resolution imaging capability of the radar. RFI signals will significantly reduce the imaging performance of the SAR system. Among them, some common wideband interference includes FM wideband interference, sinusoidal-modulated (SM) interference, and noise blocking interference.

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The FM wideband interference can be expressed as

$$\mathrm{WBI}_{\mathrm{CM}}(t,\eta)=\sum_{n=1}^{N}B_n(\eta)\exp\left(j2\pi f_{cn}t+j\pi K_n t^2\right) \tag{4}$$

where $B_n(\eta)$ represents the complex amplitude of the nth wideband interference under the ηth pulse, and K_n and f_{cn} represent the modulation frequency and center frequency of the nth wideband interference, respectively.

The SM interference is expressed as

$$\mathrm{WBI}_{\mathrm{SM}}(t,\eta)=\sum_{k=1}^{K}U_k\exp\left[j\beta_k\sin(2\pi f_{ck}t+\varphi_k)\right] \tag{5}$$

where β_k and φ_k, respectively, represent the modulation index and initial phase of the kth interference at azimuth sampling η, and f_{ck} represents the center frequency of the kth wideband interference. The characteristic of this interference is that the interference is distributed in a sinusoidal form in the time-frequency domain and causes interference to the MWP-SAR signal.

Noise blocking interference is obtained by adding a rectangular window to Gaussian white noise in the frequency domain. Its spectral function is a non-zero constant within a certain bandwidth and zero outside the bandwidth. Its power spectrum can be expressed as

$$G(f)=\begin{cases}a, & |f-f_v|\le\dfrac{B}{2}\\ 0, & \text{otherwise}\end{cases} \tag{6}$$

where B represents the interference bandwidth. f_v represents the frequency band of the interfering signal. That is, for the echo of a pulse, the signal received by the receiver may be composed of multiple interference signals. The interference signal is an additive signal to the SAR echo, that is, the echo signal can be expressed as

$$S_r(t)=S_t(t)+\mathrm{WBI}_{\mathrm{CM}}(t)+\mathrm{WBI}_{\mathrm{SM}}(t)+\Psi(G(f)) \tag{7}$$

where $\Psi(\cdot)$ represents the transformation that maps the power spectrum to the time-domain signal. This formula indicates that there are N FM wideband interferences in each pulse, K SM interferences, and noise blocking interference. Different interference signals are staggered and aliased in the time-frequency domain in the echo. Due to the presence of noise blocking interference, the interference does not have obvious sparse and low-rank characteristics compared to the echo. Therefore, it is obviously not feasible to use traditional interference detection and elimination algorithms for $S_r(t)$.

III. SRR Anti-Interference Imaging Framework

In this section, the proposed SRR algorithm architecture will be introduced in detail. This section is divided into three sub-sections and discusses how to sense interference, how to remove interference, and how to recover the signal spectrum. The flowchart of its algorithm processing is given in Fig. 1.

A. Interference Detection Model

Given that all interference forms analyzed in the previous section adversely affect image processing, the proposed SRR architecture necessitates the detection and elimination of these forms. Consequently, the problem can be effectively addressed through a binary detection approach. Additionally, the strong power spectral density of the mentioned interferences in the frequency domain motivates the development of a detection model in this domain for improved probability of success.

Therefore, the Fourier transform of the echo signal in (7) is expressed as

$$\Im_{r_n}(f_r)=\Psi\left(S_{r_n}(t)\right) \tag{8}$$

where $\Im_{r_n}(f_r)$ represents the Fourier transform of the nth pulse echo, and f_r represents the range sampling frequency. Assuming that the echo matrix collected by MWP-SAR at one time is $N\times M$ dimensional, the matrix discretization will be used and can be expressed as $\Im_r(n,m)=[\Im_{r_1}(m),\ldots,\Im_{r_N}(m)]^T, n=1,\ldots,N, m=1,\ldots,M$. This means that for a SAR imaging echo, there are N pulse sequences that need to be interfered with and sensed, and the length of each pulse sequence is M. Since the interference to SAR is considered to be in the frequency domain, interference detection can be performed on each pulse individually during interference sensing. Combined with the signal processing model of the spectrum sensing algorithm, the spectrum sensing model of the nth pulse is abstracted as

$$\begin{aligned}H_0:&\quad \Im_{r_n}(m)=x_n(m)+v_n(m)\\ H_1:&\quad \Im_{r_n}(m)=x_n(m)+\xi_n(m)+v_n(m)\end{aligned} \tag{9}$$

where $x_n(m)$ represents the echo signal emitted by MWP-SAR in the nth pulse. Additionally, $\xi_n(m)$ is used to represent other sources or interference present in the same pulse, while $v_n(m)$ characterizes the noise vector specific to the nth pulse. Utilizing the observed samples and adhering to the fundamental principles of spectrum sensing, interference detection is achieved through the statistical parameter T calculated from the observation samples.

Given the limitation that the spatial spectrum cannot be acquired in advance, the objective is to optimize the probability of correctly detecting interference. This optimization goal can be expressed as follows:

$$\begin{aligned}\max_T P_d&=\int_{\gamma}^{\infty}f_{T|H_1}(t)dt\\ \text{s.t. } P_f&=\int_{\gamma}^{\infty}f_{T|H_0}(t)dt=\varphi.\end{aligned} \tag{10}$$

The judgment threshold for the statistic T is denoted by γ. The detection probability of interference, P_d, and the false alarm probability, P_f, are defined as $P_d=P\{T>\gamma|H_1\}$ and $P_f=P\{T>\gamma|H_0\}$, respectively. The lowest acceptable probability value in interference detection is represented by φ. The test status under the assumption H_i is denoted by $T|H_i$. The detection probability under the hypothesis H_1 is represented by $f_{T|H_i}(\cdot)$.

In conventional algorithms, the sensing of narrowband interference typically outputs binary labels (0 or 1) indicating the

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TABLE I
HYPERPARAMETERS OF THE PROPOSED NETWORK

Hyperparameter	Value
Kernel size per Conv layer	30 & 60
Filter size	9
Stride	1
Padding	4
Cells per LSTM layer	128
Neurons per Dense layer	128& 64& 2
Batch size	2048
Dropout ratio	0.2

presence or absence of interference in the sensed spectrum. However, this binary output is insufficient for interference sensing in MWP-SAR. In practical engineering applications, it is necessary to obtain which segment of the wideband signal is interfered. To address this limitation, the concept of the detection bandwidth L is introduced. During signal detection, a detector of length L performs sliding detection on the received signal.

B. Spectrum Sensing Interference Detection Network

Built upon the achievements demonstrated by convolutional long short-term memory deep neural networks (CLDNNs) in modulation recognition tasks, a comparable architecture is adapted for interference perception in this article. This choice is driven by the well-suited spatial and temporal analysis capabilities of CLDNN networks, aligning with the interference sensing tasks detailed in this study.

The architecture starts with two convolution (Conv) layers and a dense layer, followed by two long short-term memory (LSTM) layers, and finally adds two Dense layers. The Conv layer is adept at capturing spatial features critical for identifying interference patterns, while the LSTM layer is adept at processing time series data, which is critical for tracking interference dynamics over time. The integration of these layers allows the network to identify complex interference patterns that vary in space and time.

For ensuring robust classification, a softmax activation function is employed in the final Dense layer, while ReLU activation functions are applied in the preceding layers to handle nonlinear data relationships. Dropout is deliberately incorporated after each layer to prevent overfitting and enhance the generalization ability of the model. The optimization of the network's hyperparameters is performed through extensive cross-validation to ensure optimal performance in the interference detection task, with details provided in Table I. Simultaneously, the network structure is depicted in Fig. 2.

The network comprises both an offline learning network and an online testing network. Unlike traditional spectrum sensing networks that can only furnish binary feedback indicating the presence or absence of interference on the receiving channel, the proposed network engages in offline training to develop a detector of length L. During real-time detection, the network employs sliding window processing on the received signal. This innovative approach significantly enhances the adaptability of the network in the context of MWP-SAR. Subsequently,

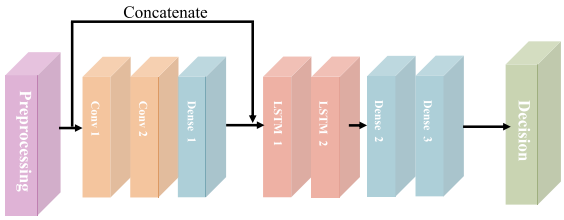


Fig. 2. Network architecture of interference detection network.

a detailed description is provided for each component of the network.

1) Offline Training Network: In the offline training network of the DNN, the initial step involves constructing the dataset. A detector with a detection length L is envisioned, wherein each sample is a complex number of length L. Simultaneously, considering the labeling of each sample, the sample set can be represented as follows:

$$\{X, Y, D\} = \left\{\left(x^{(1)}, y^{(1)}, d^{(1)}\right), \ldots, \left(x^{(K)}, y^{(K)}, d^{(K)}\right)\right\}. \quad (11)$$

Among them, $(x^{(k)}, y^{(k)}, d^{(k)})$ represents the set of kth samples, $x^{(k)}$ represents the real part of the echo signal with length L, and $y^{(k)}$ represents the imaginary part of the echo signal with length L. They are all $1 \times L$ vectors. $x^{(k)}$ and $y^{(k)}$ together form the input to the network. $d^{(k)}$ represents the label of the kth sample, $d^{(k)} = 0$ and $d^{(k)} = 1$ represent the two criteria of H_0 and H_1, respectively.

Therefore, after passing through the network, the output can be written as a 2×1 score vector

$$h_\theta\left(x^{(k)}, y^{(k)}\right) = \begin{bmatrix} h_{\theta|H_1}\left(x^{(k)}, y^{(k)}\right) \\ h_{\theta|H_0}\left(x^{(k)}, y^{(k)}\right) \end{bmatrix} \quad (12)$$

with

$$h_{\theta|H_0}\left(x^{(k)}, y^{(k)}\right) + h_{\theta|H_1}\left(x^{(k)}, y^{(k)}\right) = 1. \quad (13)$$

Among them, $h_\theta(\cdot)$ and $h_{\theta|H_i}(\cdot)$, respectively, represent the expected distribution of the output of the network under the parameter θ parameter and the output expectation under the decision H_1.

Therefore, we have the following probability expressions of two hypotheses:

$$\begin{aligned} H_1: &\quad P\left(d^{(k)} = 1 | x^{(k)}, y^{(k)}; \theta\right) = h_{\theta|H_1}\left(x^{(k)}, y^{(k)}\right) \\ H_0: &\quad P\left(d^{(k)} = 0 | x^{(k)}, y^{(k)}; \theta\right) = h_{\theta|H_0}\left(x^{(k)}, y^{(k)}\right). \end{aligned} \quad (14)$$

Based on this, the goal of the DNN training is to maximize the likelihood

$$J(\theta) = -\frac{1}{K}\sum_{k=1}^{K} d^{(k)} \log h_\theta\left(y^{(k)}\right) + \left(1 - d^{(k)}\right) \log\left(1 - h_\theta\left(y^{(k)}\right)\right). \quad (15)$$

Therefore, the solution to the optimal parameters can be expressed as

$$\theta^* = \arg\max_\theta P\left(d \middle| x^{(k)}, y^{(k)}; \theta\right). \quad (16)$$

Note that θ^* is the optimal estimate under the maximum a posteriori (MAP) criterion.

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TABLE II
DETECTION PROBABILITIES OF DIFFERENT NETWORKS FOR DIFFERENT INTERFERENCE SETS

Interference type	training data set	CNN	DNN	LSTM	Proposed
Set 1	Train	99.27%	98.92%	99.33%	**99.90%**
	Test	95.20%	88.49%	83.60%	**98.26%**
Set 2	Train	96.93%	93.97%	91.70%	**98.31%**
	Test	95.32%	87.46%	83.30%	**97.11%**
Set 3	Train	93.83%	83.33%	80.53%	**96.44%**
	Test	91.76%	81.16%	80.38%	**93.36%**

2) Online Detection Network: After the trained interference detection network has been obtained, the MWP-SAR data can be quickly processed in a streamlined manner. To maximize PD for a given PFA, the following criterion is applied:

$$L_{\mathrm{DNN}}(x,y)=\frac{h_{\theta^*|H_1}(x,y)}{h_{\theta^*|H_0}(x,y)}\cdot\frac{P(H_0)}{P(H_1)}=\frac{h_{\theta^*|H_1}(x,y)}{h_{\theta^*|H_0}(x,y)}\gtrless\gamma. \tag{17}$$

Among them, γ represents the discrimination threshold that needs to be used during detection. The threshold could be derived from the false alarm constraint. As shown in Fig. 1, the data sequence of the pipeline input test network is

$$\left\{\hat{X}^{(k)},\hat{Y}^{(k)}\right\}=\left\{\left(\hat{x}_1^{(k)},\hat{y}_1^{(k)}\right),\ldots,\left(\hat{x}_L^{(k)},\hat{y}_L^{(k)}\right)\right\}. \tag{18}$$

The online judgment is available

$$\begin{aligned}L_{\mathrm{DNN}}\left(\hat{X}^{(k)},\hat{Y}^{(k)}\right)&=\frac{h_{\theta^*|H_1}\left(\hat{X}^{(k)},\hat{Y}^{(k)}\right)}{h_{\theta^*|H_0}\left(\hat{X}^{(k)},\hat{Y}^{(k)}\right)}\cdot\frac{P(H_0)}{P(H_1)}\\&=\frac{h_{\theta^*|H_1}\left(\hat{X}^{(k)},\hat{Y}^{(k)}\right)}{h_{\theta^*|H_0}\left(\hat{X}^{(k)},\hat{Y}^{(k)}\right)}\gtrless\gamma.\end{aligned} \tag{19}$$

Once the test statistic is obtained, a decision can be made by comparing it with a preset threshold. After determination through the online interference sensing network, the interfered components in the spectrum can be accurately labeled. In order to prevent interference signals from interfering with subsequent imaging, the interference part of the echo frequency-domain model represented by (7) is set to zero. The echo of the interference signal is eliminated, which is expressed by the following formula:

$$\begin{aligned}\tilde{\Im}_r(n,m)&=\left[\tilde{\Im}_{r_1}(m),\ldots,\tilde{\Im}_{r_N}(m)\right]^T\\&n=1,\ldots,N,\quad m=1,\ldots,M.\end{aligned} \tag{20}$$

C. Recovery of Missing Spectrum

Next, for the signal after removing the interference, it is necessary to restore the missing signal components to ensure the resolution of subsequent imaging processing. Therefore, in this section, an efficient spectrum reconstruction algorithm will be proposed to improve the imaging processing accuracy of MWP-SAR.

1) Spectrum Missing Model: Due to the echo matrix, the spectrum loss of each pulse signal can be restored. We consider that the pulse signal to be recovered is $\tilde{\Im}_r(m)=[\tilde{\Im}_r(1),\ldots,\tilde{\Im}_r(M)],m=1,\ldots,M$. Among them, part of the spectrum components are zero-set frequency bands after interference is eliminated. The matrix completion algorithm not only requires the matrix to be processed to be a low-rank matrix, but also has certain restrictions on the observation values and observation methods. The matrix $\tilde{\Im}_r(m)$ that needs to be restored cannot be missing entire rows and columns.

In order to solve this problem, the vector can be transformed into a low-rank Toeplitz matrix through the linear transformation operator $T:C^{M\times1}\rightarrow C^{n_1\times n_2}$. For vector $\tilde{\Im}_r(m)=[\tilde{\Im}_r(1),\ldots,\tilde{\Im}_r(M)],m=1,\ldots,M$, set the interval d, and its Toeplitz matrix is defined in (21), as shown at the bottom of the next page. The changes in the rank of the data matrix before and after transformation are shown in Fig. 3. It can be seen that the Toeplitz transform effectively increases the rank of the data matrix, laying the foundation for subsequent signal recovery.

In this article, the aim is to have the best estimate of $\hat{S}_{\mathrm{est}}$ for this matrix. Therefore, this article establishes the spectrum recovery problem as approximating the rank function with the nuclear norm, and performing convex relaxation on the rank function to obtain the spectrum recovery model

$$\begin{aligned}&\min_{\hat{S}_{\mathrm{est}}}\left\|\hat{S}_{\mathrm{est}}\right\|_*\\&\text{s.t. }P_\Omega\left(\hat{S}_{\mathrm{est}}\right)=P_\Omega\left(T\left(\tilde{\Im}_r(m)\right)\right).\end{aligned} \tag{22}$$

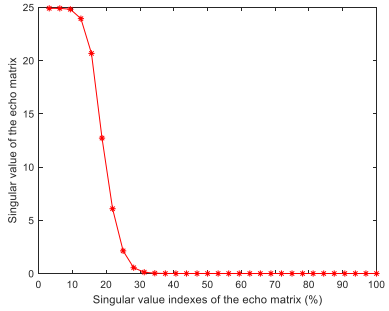
In (22), $T(\tilde{\Im}_r(m))$ represents the observation matrix corresponding to the image under restoration, Ω denotes the set of positions associated with known elements of the matrix $T(\tilde{\Im}_r(m))$, and $P_\Omega(\cdot)$ stands for the linear projection operator. However, for large matrices, minimizing nuclear norm models will incur huge complexity due to the requirement of singular value decomposition in optimization iterations. At the same time, for the problems described in the above formula, the Schatten-p norm problem is usually used for equivalent replacement. However, the Schatten-p norm is not a convex function globally, so it may easily fall into a local optimum during optimization.

Consequently, this article adopts the FGSR model to restore large-scale low-rank matrices. This model incorporates a factor group sparse regularizer to relax the number of non-zero columns in matrix decomposition, thereby constructing a new regularizer. Its notable advantage lies in providing a more stringent approximation to the rank function, coupled with robust initialization and convergence effects. This enables

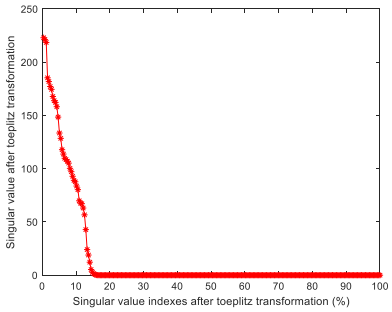
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TABLE III
COMPARISONS OF IMAGING INDICATORS PROCESSED BY DIFFERENT ANTI-INTERFERENCE ALGORITHMS

Interference type	Image index	Frequency domain filtering	RPCA	GoDec	Proposed
Set 1	SSIM	0.5228	7.21E-16	0.0966	**0.9092**
	Image Entropy	8.0397	11.6498	10.2076	**7.2218**
	SNR	13.9062	0.7519	5.8841	**17.0209**
Set 2	SSIM	0.5442	7.21E-16	0.0925	**0.9661**
	Image Entropy	7.8592	11.4668	9.8264	**7.1976**
	SNR	14.9434	1.0480	6.6573	**17.1328**
Set 3	SSIM	0.3173	2.77E-4	1.63E-4	**0.7841**
	Image Entropy	8.5841	11.2196	11.2648	**7.2252**
	SNR	12.4840	2.0597	2.3059	**17.0033**



(a)



(b)

Fig. 3. Analysis of low-rank characteristics of Toeplitz transform. (a) Singular value distribution before Toeplitz transform. (b) Singular value distribution after Toeplitz transform.

non-zero columns to rapidly converge and stabilize, yielding accurate iterative results with minimal computational overhead. Building upon the FGSR framework, the problem (22) can be reformulated as

$$\frac{2}{3\alpha^{1/3}}\min_{A,B}\|A\|_{2,1}+\frac{\alpha}{2}\|B\|_F^2$$
$$\text{s.t. } P_\Omega\left(\hat{S}_{\text{est}}\right)=P_\Omega\left(T\left(\tilde{\Im}_r(m)\right)\right),\quad \hat{S}_{\text{est}}=AB. \tag{23}$$

Among them, α is a regularization parameter. Assume that the singular value decomposition of matrix $\hat{S}_{\text{est}}$ is: $\hat{S}_{\text{est}}=USV^H$, then, $A=\alpha^{1/3}US^{2/3}$, $B=\alpha^{-1/3}S^{1/3}V^H$. $\|A\|_{2,1}=\sum_{j=1}^{d}a_j$ represents the sum of the non-zero column two norms of the matrix. The augmented Lagrangian function of the above problem is

$$L\left(A,B,Y,\hat{S}_{\text{est}}\right)=\|A\|_{2,1}+\frac{\alpha}{2}\|B\|_F^2 + \left\langle Y,\hat{S}_{\text{est}}-AB\right\rangle+\frac{\rho}{2}\left\|\hat{S}_{\text{est}}-AB\right\|_F^2. \tag{24}$$

Among them, Y is the Lagrange multiplier and ρ is a positive penalty factor. To solve the above problems using the ADMM algorithm framework, we can divide them into the following sub-problems.

2) Iteratively Solve A^{k+1}: Since $\|A\|_{2,1}$ represents the sum of the 2-norms of the non-zero columns of the matrix. Therefore, each column vector of the matrix can be soft-thresholded to achieve column sparsity, that is, only some column vectors are retained and other column vectors are set to zero. This can reduce the rank of the matrix, reduce redundant information, and improve decomposition efficiency and accuracy. Then the update of A^{k+1} can be expressed as

$$A^{k+1}=\underset{A}{\text{argmin}}\,L\left(A,B^k,Y^k,\hat{S}_{\text{est}}^k\right) =\Psi\left[A^k+\rho^k\left(\hat{S}_{\text{est}}^k-A^kB^k+\frac{Y^k}{\rho^k}\right)\left(B^k\right)^H\Big/\tau,\frac{1}{\tau}\right] \tag{25}$$

where

$$\Psi\left(W,\lambda\right)=\begin{bmatrix}\Phi\left(W_1,\lambda\right)\\ \Phi\left(W_2,\lambda\right)\\ \vdots\\ \Phi\left(W_n,\lambda\right)\end{bmatrix} \tag{26}$$

$$T\left(\tilde{\Im}_r(m)\right)\triangleq\begin{bmatrix}\tilde{\Im}_r(1) & \tilde{\Im}_r(M-d+1) & \cdots & \tilde{\Im}_r(M-1) & \tilde{\Im}_r(M)\\ \tilde{\Im}_r(2) & \tilde{\Im}_r(1) & \cdots & \tilde{\Im}_r(M-2) & \tilde{\Im}_r(M-1)\\ \vdots & \vdots & \ddots & \vdots & \vdots\\ \tilde{\Im}_r(M-d-1) & \tilde{\Im}_r(M-d-2) & \cdots & \tilde{\Im}_r(1) & \tilde{\Im}_r(M-d+1)\\ \tilde{\Im}_r(M-d) & \tilde{\Im}_r(M-d-1) & \cdots & \tilde{\Im}_r(2) & \tilde{\Im}_r(1)\end{bmatrix} \tag{21}$$

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and

$$\Phi(W_i, \lambda) = \begin{cases} \dfrac{\|W_i\|_2 - \lambda}{\|W_i\|_2} W_i, & \text{if } \|W_i\|_2 > \lambda \\ 0, & \text{otherwise} \end{cases} \tag{27}$$

where W_i represents the ith column vector in A.

If $\|W_i\|_2$ is greater than the threshold λ, then shrink it. If $\|W_i\|_2$ is less than or equal to the threshold λ, then set the column to 0, eliminate the contribution of the column vector. The threshold λ here is represented by $1/\tau$. Among them, $\tau = \rho^k \|B^k (B^k)^H\|_2$.

Algorithm 1 FGSR-ADMM Spectrum Recovery Algorithm

Input:

Observation matrix to be recovered $T(\tilde{\Im}_r(m))$, ADMM maximum iteration number K, algorithm intermediate parameter set α, β, ρ^0, d;

initialization:

$A^0 = \text{rand}(m, d)$, $B^0 = \text{rand}(d, n)$, $Y^0 = 0$, $\hat{S}^0_{est} = T(\tilde{\Im}_r(m))$, $k = 0$;

Output:

1: Step 1: Update variable A^{k+1} according to the algorithm flow and soft threshold algorithm in the first section of this paper;

2: Step 2: Update B^{k+1}.

$$B^{k+1} = \left(\alpha I + \rho^k \left(A^{k+1}\right)^H A^{k+1}\right)^{-1} \left(\rho^k \left(\left(A^{k+1}\right)^H \left(\hat{S}^k_{est} + \frac{Y^k}{\rho^k}\right)\right)\right);$$

3: Step 3: Update $\hat{S}^{k+1}_{est}$.

$$\hat{S}^{k+1}_{est} = A^{k+1} B^{k+1} - \frac{Y^k}{\rho^k};$$

4: Step 4: Update Y^{k+1}.

$$Y^{k+1} = Y^k + \rho^k \left(\hat{S}^{k+1}_{est} - A^{k+1} B^{k+1}\right);$$

5: Step 5: $\rho^{k+1} = \beta \rho^k$.

6: Step 6: Determine whether the number of iterations reaches the maximum, if not satisfied, go to Step 1, and set $k = k + 1$; if satisfied, output $\hat{S}^{k+1}_{est}$.

3) Iteratively Solve for B^{k+1} and $\hat{S}^{k+1}_{est}$: Since the objective function constructed above is a convex function, B^* is the optimal solution if and only if

$$\nabla L_B(A, B, Y, \hat{S}_{\text{est}}) = 0 \Rightarrow \alpha B^* - Y A^H - \rho A^H (\hat{S}_{\text{est}} - A B^*) = 0. \tag{28}$$

Therefore, the least square method can be used to simplify the above equation and obtain the optimal estimate of B^*

$$B^* = \left(\alpha I + \rho (A)^H A\right)^{-1} \left(\rho \left((A)^H \left(\hat{S}_{\text{est}} + \frac{Y}{\rho}\right)\right)\right). \tag{29}$$

In the same way, the partial derivative of the objective function can be obtained to give the minimum value of $\hat{S}^*_{\text{est}}$, that is,

$$\nabla L_{\hat{S}_{\text{est}}}(A, B, Y, \hat{S}_{\text{est}}) = 0 \Rightarrow Y + \rho (\hat{S}^*_{\text{est}} - A B) = 0. \tag{30}$$

The concise expression of the optimal value of $\hat{S}^*_{\text{est}}$ can be obtained as

$$\hat{S}^*_{\text{est}} = A B - \frac{Y}{\rho}. \tag{31}$$

Therefore, in summary, the solution process of FGSR-ADMM can be represented by Algorithm 1.

IV. Building Network Performance

In order to prove the effectiveness of the above algorithm, the performance of the detection network needs to be analyzed to ensure the effective operation of the SRR imaging architecture.

In this part, a large number of disturbed datasets are first generated to meet the learning requirements of the network. During the data generation process, the intricate spectrum environment typical of MWP radar operation is taken into account. Three forms of interference are simulated. The first involves broadband suppression interference originating from a single interference source, where the interference bandwidth constitutes 40% of the signal frequency band. The simulated MWP radar experiences robust suppression and interference from this single source. The second type encompasses joint interference from multiple sources within the same bandwidth, with the interference frequency band representing 25% of the signal. In the simulated MWP working environment, there are additional narrowband radar signals or communication signals. The third type involves interference induced by multiple sources distributed across different frequency bands, with the interfered frequency band making up 50% of the signal. The generated time-frequency diagrams of the three types of interference are shown in Fig. 4.

For each signal type, both the center frequency and frequency modulation slope of the interference are randomly determined during generation. This approach ensures a comprehensive coverage of interference types within the generated training set. A total of 10 000 samples are generated for each type of interference. The resulting samples are then categorized into training and test sets, with the training set comprising 70% of the samples and the test set containing the remaining 30%. In the simulation, the Net proposed in this article is implemented based on the Pytorch framework on the Windows 11 operating system. The Adam optimizer is used to train the network.

In this article, in order to demonstrate the advantages of the proposed detection network, we compare a variety of existing spectrum sensing algorithms, including CNN networks [30], DNN [31] networks and LSTM networks [32]. At the same time, the detection probabilities of various sensing networks for different interference sets are evaluated. The effects of each network in the training set and test set are analyzed, and the results are shown in Fig. 5. The specific detection probability values are shown in Table II. It can be seen from the detection results that in the face of complex interference forms, the perception proposed network can still obtain a high detection probability. Compared with traditional deep networks, the network in this article not only considers the time and frequency characteristics of the signal

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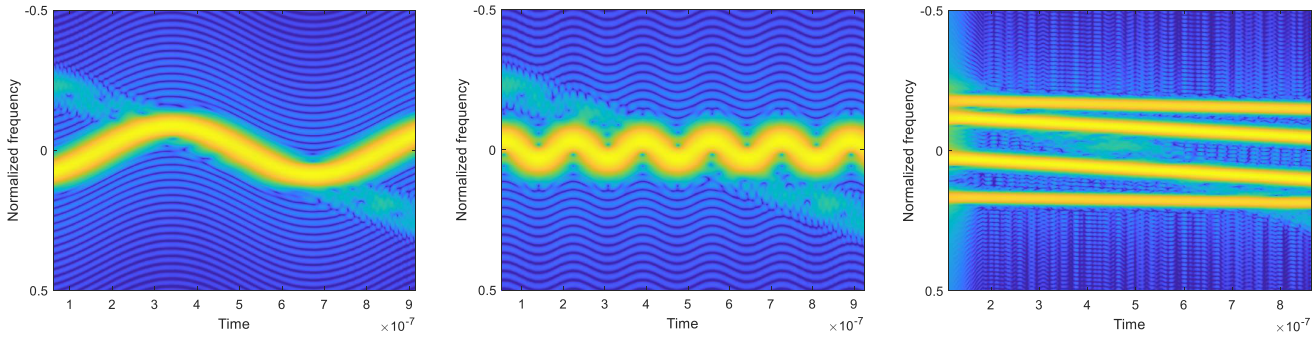


Fig. 4. Time-frequency diagrams of different interfering signals. (a) Interference dataset 1. (b) Interference dataset 2. (c) Interference dataset 3.

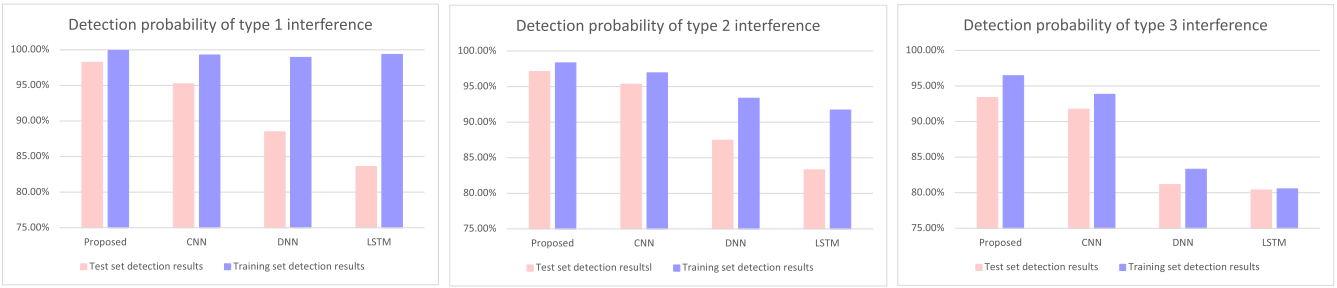


Fig. 5. Detection effects of different existing detection networks on different interference data. (a) Interference dataset 1. (b) Interference dataset 2. (c) Interference dataset 3.

TABLE IV
COMPARISONS OF IMAGING INDICATORS PROCESSED BY DIFFERENT ANTI-INTERFERENCE ALGORITHMS

Interference type	Image index	Frequency domain filtering	RPCA	GoDec	Proposed
Set 1	Contrast	384.1549	184.1511	345.0285	**392.3502**
	Image Entropy	6.3481	10.6268	10.7711	**4.5410**
	SNR	32.2931	11.7050	12.5644	**36.7181**
Set 2	Contrast	387.5589	184.7272	347.6547	**392.4892**
	Image Entropy	6.1376	10.5132	10.9050	**4.8866**
	SNR	33.3857	11.5472	11.1482	**36.1378**
Set 3	Contrast	383.6250	185.2091	351.2988	**392.5642**
	Image Entropy	6.8237	10.4800	10.4593	**4.6805**
	SNR	31.6449	9.2544	8.6862	**36.0620**

at the same time, but also uses small sub-band pipeline scanning to accurately detect interference in ultra-wideband signals.

The network's perceptual performance with respect to the tolerable false alarm is studied, as shown in Fig. 6. Since the proposed algorithm will remove the interfered spectrum when interference is detected, the false alarm spectrum will cause uninterrupted components in the signal to be removed. However, the subsequent recovery algorithm can recover signals with continuous missing up to 50%. Therefore, this architecture can tolerate a higher false alarm rate during interference detection. In this case, this article proposes that the probability of detecting the network will be infinitely close to 100%.

Further, this article analyzes the learning performance of the network. The variation pattern of the detection probability of each network with the training sample batch is shown in Fig. 7. It can be seen that the proposed network can still achieve good performance with small amount of training samples. This facilitates the deployment of the proposed network on the hardware platform.

The architecture of our proposed deep learning network is designed to achieve superior interference detection accuracy by integrating CNNs for spatial analysis and LSTMs for capturing temporal dynamics. This integrated approach allows for real-time adaptation to evolving interference scenarios, a critical capability for operational robustness in MWP-SAR applications.

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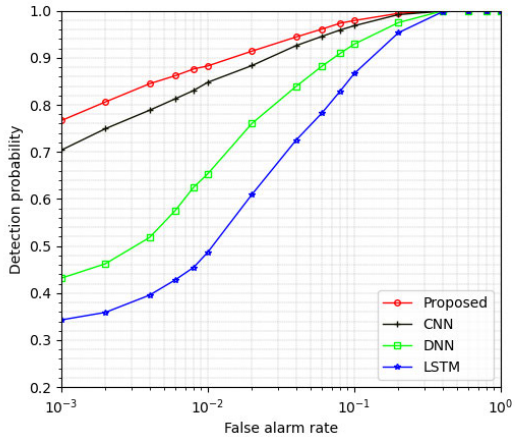


Fig. 6. Detection probability changes versus the false alarm probability.

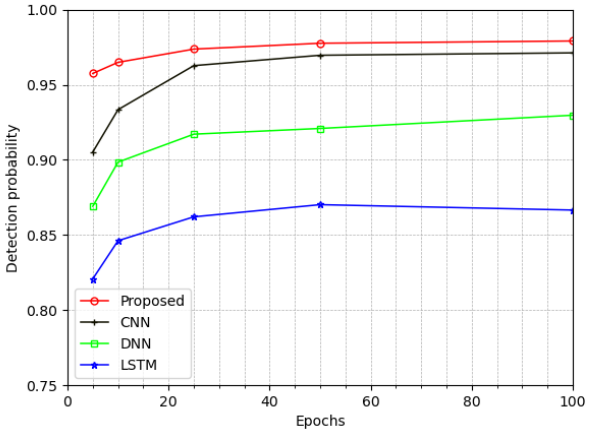


Fig. 7. Learning speed of different networks.

V. Simulation and Measured Data Processing

In evaluating the effectiveness and advantages of the algorithm proposed in this article in comparison to existing anti-interference algorithms, two segments have been devised: numerical simulation and processing of measured data. To ensure the authenticity of the simulation data, the numerical simulation parameters have been aligned with those utilized in the microwave photonic system described in the literature [33]. The SRR imaging architecture proposed in this article will be directly compared with a variety of mature anti-interference SAR imaging algorithms to reflect its advantages.

A. Numerical Simulation

The simulation in the previous section verified that the proposed interference-aware network proposed in this article can detect various interferences skillfully and accurately, and has higher detection probability and performance compared with traditional interference detection networks. Since each pulse signal is independent of each other. Therefore, in this section, we illustrate the effectiveness of the proposed signal recovery algorithm with a set of samples from the study. This sample set encompasses three interference scenarios. Fig. 9 portrays the evolution of the signal spectrum processed by the

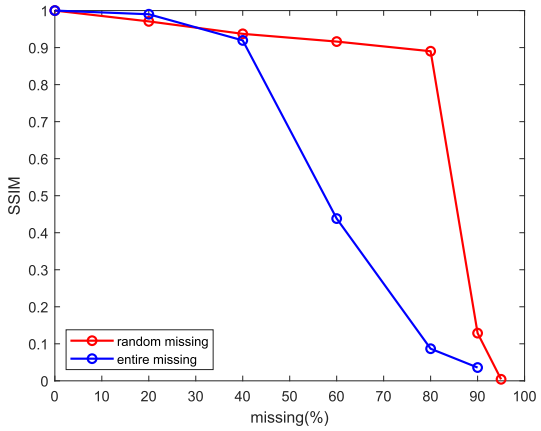


Fig. 8. Recovery effects of spectrum recovery algorithms under different interference types.

algorithm. The first row of Fig. 9 displays the spectrum of the interfered signal. Leveraging the aforementioned network, accurate detection of the interference spectrum's location is achieved. Consequently, the second row illustrates the scenario after removing the interference spectrum under different interference conditions. The third row showcases the recovery efficacy of the proposed signal recovery algorithm for the missing signal. Remarkably, this algorithm exhibits proficient restoration of diverse forms of spectrum loss. However, due to the space limit of the article, they cannot be listed here.

This article also analyzes the recovery ability of the spectrum recovery part of the proposed algorithm for different missing forms, as shown in Fig. 8. The vertical axis in the figure represents the Structure Similarity Index Measure (SSIM) between the restored signal and the original spectrum, and the horizontal axis represents the lack of signal. It can be seen that for the entire spectrum missing, the algorithm proposed in this article can effectively recover when the missing degree is not greater than 60%. For random missing in the spectrum, this article proposes an algorithm that can effectively restore the signal when the missing degree is no more than 80%. This also further improves the robustness of the SRR imaging architecture.

Furthermore, a thorough analysis of the overall algorithmic performance is carried out to showcase the superiority of the algorithm proposed in this article. The performance of this algorithm will be compared with various existing anti-interference algorithms, including the frequency-domain filtering algorithm, RPCA [34], and GoDec [35]. The interference suppression effects of each algorithm are depicted in Fig. 10. In this simulation, the ultra-wideband point target data echo is generated through MATLAB software, and the above-mentioned algorithm is used to process the signal.

From the point target simulation, it can be observed that the various forms of interference signals added can be better suppressed by the proposed algorithm, leading to interference-free imaging results. However, traditional anti-interference algorithms are unable to suppress large-bandwidth interference. This article gives a comparison of imaging image indicators processed by different anti-interference algorithms,

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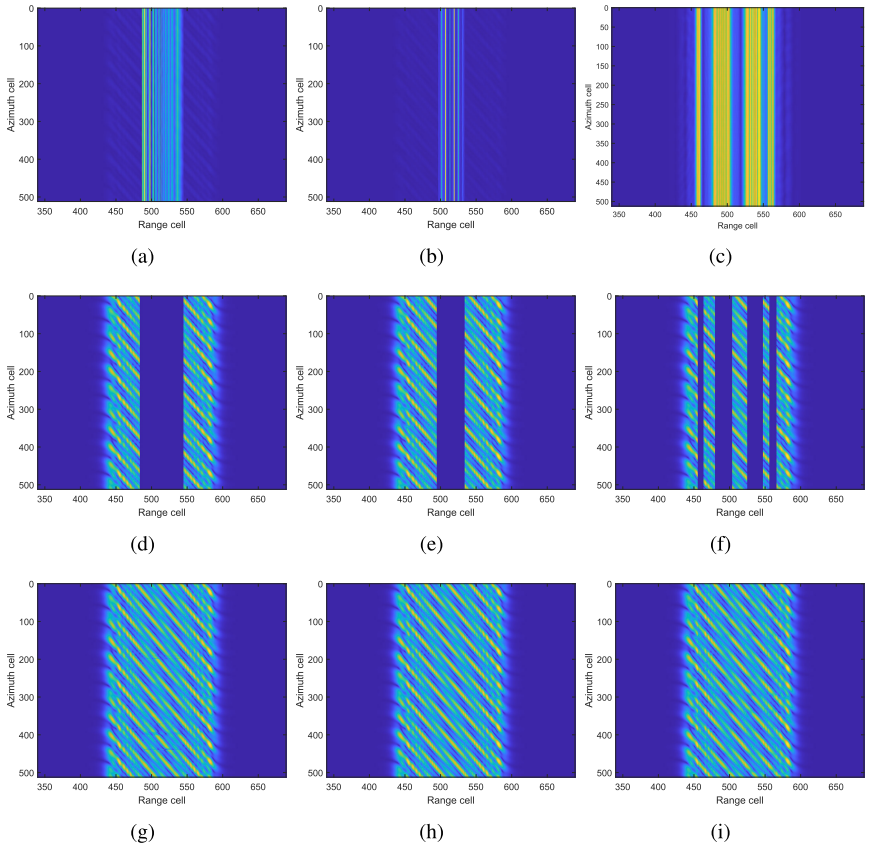


Fig. 9. Spectrum diagrams of simulation data at different processing stages after adding different interferences. (a) Add the echo spectrum of the first interference. (b) Add the echo spectrum of the second interference. (c) Add the echo spectrum of the third interference. (d) Spectrum sensing signal spectrum after removing the first interference. (e) Spectrum sensing signal spectrum after removing the second interference. (f) Spectrum sensing signal spectrum after removing the third interference. (g) Algorithm restores the first missing spectrum. (h) Algorithm restores the second missing spectrum. (i) Algorithm restores the third missing spectrum.

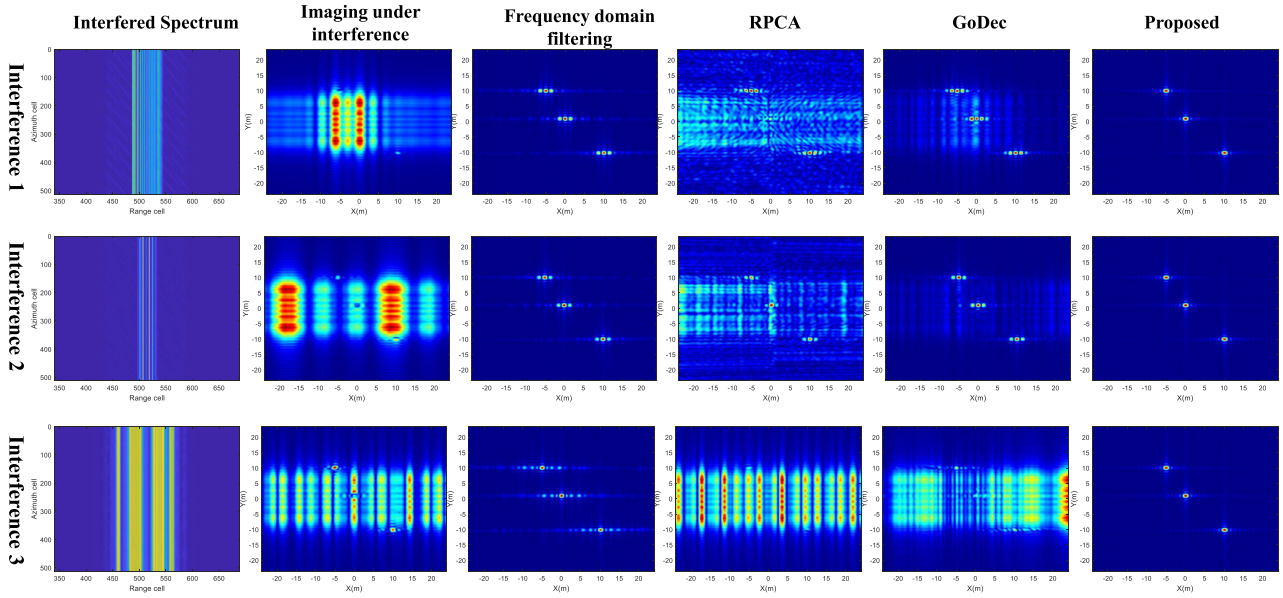


Fig. 10. Comparisons of performance between the proposed algorithm and existing anti-interference algorithms.

as shown in Table III. The index SSIM is calculated from the imaging results without interference and the imaging after anti-interference. This table can also visually reflect the advantages of this algorithm.

B. Measured Data Processing

The airborne data used in the current verification can be collected by referring to the system shown in the literature [36]. The MWP-SAR system is installed on the Cessna 208 B

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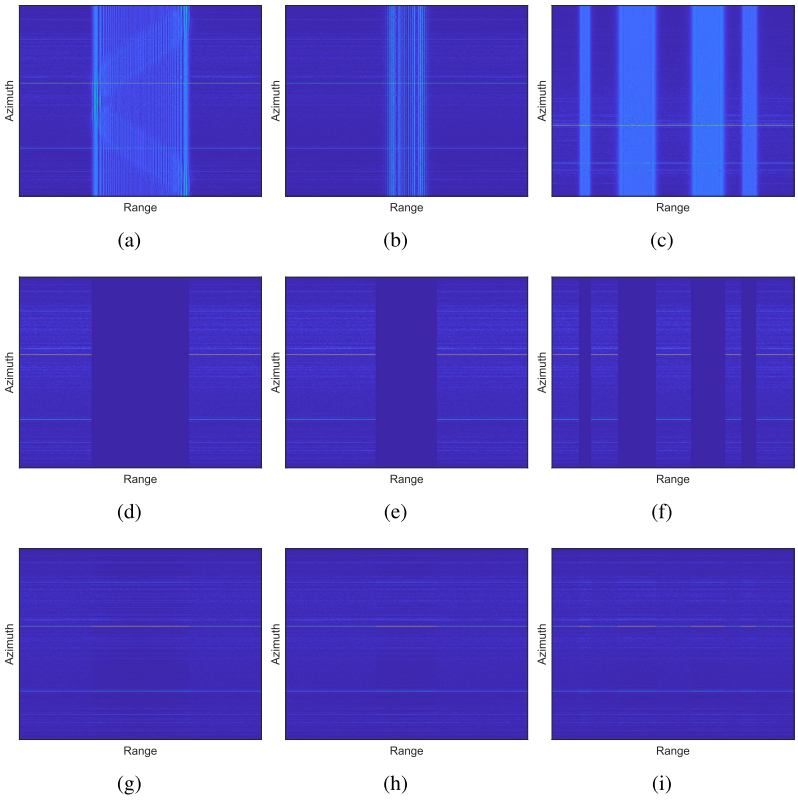


Fig. 11. Spectrum diagrams of measured data at different processing stages after adding different interferences. (a) Add the echo spectrum of the first interference. (b) Add the echo spectrum of the second interference. (c) Add the echo spectrum of the third interference. (d) Spectrum sensing signal spectrum after removing the first interference. (e) Spectrum sensing signal spectrum after removing the second interference. (f) Spectrum sensing signal spectrum after removing the third interference. (g) Algorithm restores the first missing spectrum. (h) Algorithm restores the second missing spectrum. (i) Algorithm restores the third missing spectrum.

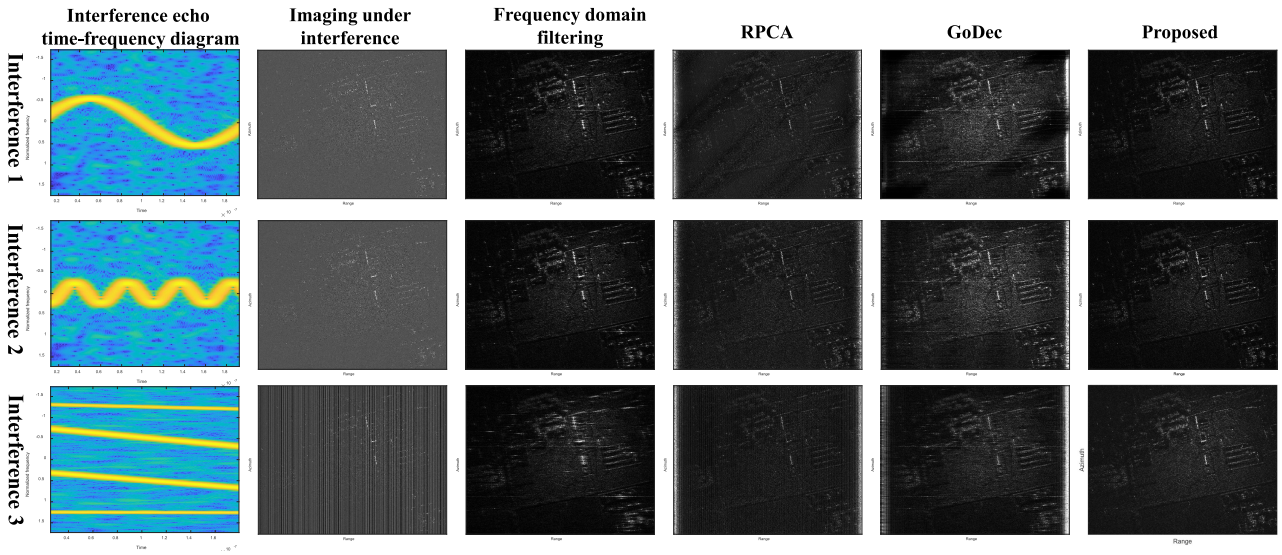


Fig. 12. Comparisons of performance between the proposed algorithm and existing anti-interference algorithms.

aircraft and horn antenna (20 W, CW) is used to transmit and receive. The system achieves centimeter-level resolution imaging of ground scenes. The SAR images used in this article are processed using the optimal aperture time-domain imaging algorithm [5]. At the same time, the MWP-SAR system uses IMU/GNSS (Applanix AP60) to collect attitude data and provide motion compensation for signal processing. The system has a center frequency of 15.14 GHz and a bandwidth of 5.72 GHz.

It is worth mentioning that there is no RFI in the actual admitted echo. Before signal processing, the interference samples generated for perception training are added to the measured echo to simulate a real complex electromagnetic environment. The specific form of interference is described

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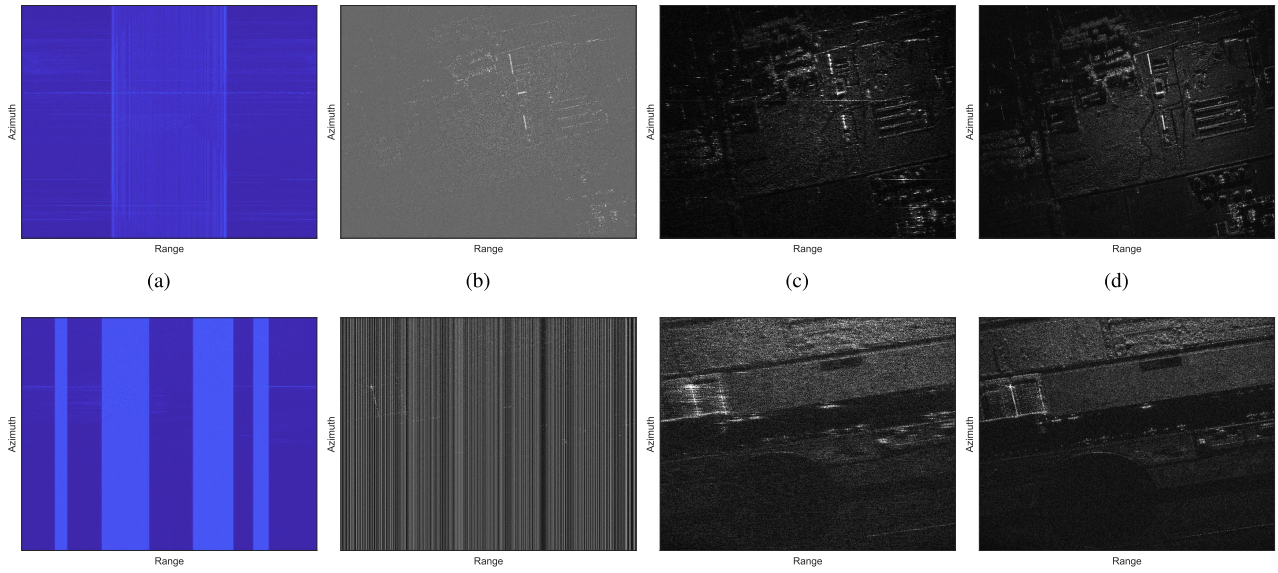


Fig. 13. Use measured data (two scenarios) to verify the SRR algorithm. (a) Echo range frequency domain with interference (Scenario 1). (b) Image interfered (Scenario 1). (c) After filtering out interference bands (Scenario 1). (d) Image after recovering missing spectrum (Scenario 1). (e) Echo range frequency domain with interference (Scenario 2). (f) Image interfered (Scenario 2). (g) After filtering out interference bands (Scenario 2). (h) Image after recovering missing spectrum (Scenario 2).

in Sections II and IV. and the previous section of the article.

Fig. 11 provides a detailed illustration of the signal spectrum changes throughout the signal processing procedure induced by the proposed algorithm. The employed SRR processing architecture is strictly adhered to during the processing stages. It is evident that, leveraging a high-precision sensing network and an effective missing signal recovery algorithm, the interference spectrum is initially detected with high precision by the spectrum sensing network. Subsequently, the interference spectrum is accurately eliminated. Finally, the missing signal spectrum is effectively restored.

Further, different anti-interference algorithms are used to process the echoes under different interferences. The processing results are shown in Fig. 12. From the processing results, the efficacy of the frequency-domain filter's anti-interference algorithm in entirely eliminating interference signals is evident. However, this process results in a loss of imaging signal bandwidth, leading to a reduction in imaging resolution, which is deemed unacceptable for MWP imaging radar, renowned for its ultrahigh resolution imaging capability. Conversely, the anti-interference algorithm relying on sparse-low rank and principal component decomposition struggles to effectively suppress diverse interference signal forms due to their inherent complexity. Analysis of the result indicates that while the existing interference suppression algorithm moderately enhances the imaging SNR, it fails to meet the application requirements. The proposed algorithm adeptly detects, identifies, and eliminates diverse forms of interference signals leveraging the detection network. Subsequently, the missing signal spectrum is restored. Moreover, to conduct a quantitative analysis of the proposed algorithm's performance, the outcomes are presented in Table IV, allowing for a clear and intuitive comparison of the anti-interference effects of different algorithms.

Finally, in order to intuitively display the changes in MWP-SAR images in the SRR imaging architecture, this article provides a large picture of the processing results in Fig. 13. In the echoes of different scenarios, we added two different interferences, which are interference type 1 and interference type 3 mentioned in this article. The echo after adding interference is shown in the first column of Fig. 13. Then, for different interferences, this article uses the proposed algorithm to process and can achieve better interference suppression effects, verifying the applicability of the algorithm in echoes in different scenarios. In conclusion, the proposed algorithm has demonstrated favorable outcomes in processing MWP-SAR data under diverse interferences, emphasizing the algorithm's effectiveness.

VI. Conclusion

In conclusion, this research endeavors to address the challenges posed by complex electromagnetic interference in MWP-SAR systems. An interference suppression imaging algorithm grounded in deep spectral sensing and spectrum reconstruction has been proposed. The high-resolution imaging problem of MWP-SAR in complex electromagnetic environments has been successfully addressed by the established SRR anti-interference imaging framework. Through the utilization of a deep learning network based on spectrum sensing theory, we achieve an impressive 98% interference detection probability under diverse interference scenarios. Additionally, our proposed signal reconstruction algorithm, employing Toeplitz transformation, effectively addresses the challenge of continuous spectrum loss post-interference removal. Simulation

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and real data processing have confirm the superior performance of our proposed algorithm over existing interference suppression imaging methods. The algorithm can consistently deliver ultrahigh-resolution images of the observation area, demonstrating robustness even under intense electromagnetic interference conditions.

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